

Avalon-MM SDRAM Controller IP Core

User Guide

IP core	<code>altera_avalon_mm_sdram_controller</code>
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1. About this core

1.1 Features

- **One open row per bank.** Four rows held open at once rather than one, so an access that changes bank costs no row command.
- **Read/write turnaround without a row cycle.** 0 cycles write → read, CAS+1 read → write, as the device datasheet describes.
- **Look-ahead row activation.** The row for the *next* buffered access is opened or closed while the current one is still being served.
- **Postponed refresh.** Up to `REF_MAX_PEND` refreshes are deferred to a quiet moment, then forced through.
- **Parameterised in time, not cycles.** Device timings are picoseconds; cycles are derived inside the HDL.
- **Drop-in.** Same `s1` slave, same `wire` conduit, same default address map as the core it replaces.

1.2 What it is for

Mixed and scattered traffic — which is what a CPU generates. Sequential streaming was already at 97% of the bus limit before this core existed and is not where the headroom was. See section 7.

1.3 Status

Verified in simulation, synthesised, and closing timing — but never run on a board. The core compiles through Quartus 18.1.1 Standard for the DE10-Lite's MAX 10, fits, meets a 100 MHz constraint with 0.430 ns of setup slack, and produces a bitstream. It has not been programmed into a part, so refresh and retention on silicon remain unproven: no functional model forgets. Resource and `f_MAX` figures are in section 7.4 and are reproducible.

2. Getting started

2.1 Adding the component

Add the repository to the IP search path — **Tools ► Options ► IP Search Path**, or `--search-path` on the command line — and the component appears in the IP Catalog under **Memory Interfaces and Controllers / Custom** as *Avalon-MM SDRAM Controller (per-bank rows)*.

2.2 Start from a preset

Select the preset for your part rather than typing timings in:

```
add_instance sdram altera_avalon_mm_sdram_controller
apply_preset sdram "ISSI IS42S16320D-7 - DE10-Lite 64 MByte"
```

A preset carries the geometry, the timings and the refresh figures for one device, taken from its datasheet once. Two are supplied:

Preset name	Part	Size	Banks x rows x cols x bits
ISSI IS42S16320D-7 - DE10-Lite 64 MByte	IS42S16320D-7	64 MB	4 x 8192 x 1024 x 16
ISSI IS42S16160B-7 - DE0-Nano 32 MByte	IS42S16160B-7	32 MB	4 x 8192 x 512 x 16

Only parts whose timing has been checked against a datasheet *and* exercised through `benchmark/` and the testbench are supplied; adding one is a block of XML in `altera_avalon_mm_sdram_controller.qprs`. Because a preset is a datasheet transcribed, both are held in place from two directions: `doc/tools/check_facts.py` compares each preset against the copy the benchmark and testbench carry, and the regression simulates each part's geometry and timings rather than only linting them.

Caution

There is deliberately no "generic SDRAM" preset. A timing constant that is one speed grade optimistic produces a design that passes every simulation, works on the bench, and corrupts data once the board is warm.

2.3 Connecting it

Interface	Connect to
clk, reset	Your clock source. The clock rate is read from it , so no frequency is typed in anywhere
s1	The master, or the interconnect
wire	Export to the top level, and constrain the SDRAM pins

2.4 What is checked at generation time

The component refuses, or warns about, four configurations that would otherwise build and be wrong on hardware:

Condition	Result
<code>SA_BITS</code> too small for <code>ROW_BITS</code> , or below 11	Error
<code>tRAS</code> longer than <code>tRC</code> , or <code>tRRD</code> longer than <code>tRC</code>	Error
A refresh interval the controller cannot sustain	Error
<code>ADDR_MAP</code> set to conventional	Warning — every address moves

It also reports the memory size and the exact cycle counts the HDL will derive at your clock, so they can be read against the datasheet:

```
Info: Memory: 64 MByte - 4 banks x 8192 rows x 1024 columns x 16 bits.
Info: At 143.000 MHz the HDL will use: tRC=9 tRAS=6 tRP=3 tRCD=3 tRRD=3
      tWR=3 tRFC=9 cycles, CAS=3, one refresh every 1118 cycles.
```

3. Functional description

3.1 Initialisation

After reset the controller holds `za_waitrequest` high and runs the JEDEC power-up sequence: wait `T_INIT_US`, PRECHARGE ALL, `INIT_REFS` AUTO REFRESH commands, then LOAD MODE REGISTER. Only then is a transfer accepted.

The mode register is written with burst length 1, sequential burst type, and the configured CAS latency.

3.2 Serving a transfer

One command is issued per cycle, chosen in priority order — see Figure 3 in the block-diagram document. In summary: serve the head if its row is open, close the row if it is the wrong one, open the bank if it is closed, otherwise prepare the next access.

3.3 Read/write turnaround

A READ issued at cycle T has the device driving DQ at $T+CAS$. A WRITE drives DQ in its own cycle, so the earliest safe write is $T+CAS+1$. The other direction is free — the datasheet permits write data to be immediately followed by a READ command.

This is why mixed traffic settles at about 79 MB/s rather than 194: a write/read pair costs five cycles at CAS 3, and two accesses per five cycles is 80 MB/s. That is the device, not the scheduler.

3.4 Refresh

A timer accumulates outstanding refreshes at the interval implied by `REF_ROWS` and `REF_PERIOD_MS`. They are issued when the command buffer is empty, or forced once `REF_MAX_PEND` have accumulated. A refresh precharges every bank, so all open rows are lost and reopened as needed.

3.5 Byte enables

`az_be_n` is active low and is driven onto `zs_dqm` for writes. DQM is held low at all other times, which satisfies the SDR read-output enable requirement.

4. Parameters

4.1 Geometry

Parameter	Type	Default	Range	Description
DATA_BITS	Integer	16	8, 16, 32	SDRAM data bus width. One DQM bit per byte lane
ROW_BITS	Integer	13	10:15	Row address width
COL_BITS	Integer	10	8:11	Column address width. Column bit 10 steps over A10
BANK_BITS	Integer	2	1, 2, 3	Bank address width. Also how many rows can be open at once
SA_BITS	Integer	13	11:15	Address pins on the device. At least ROW_BITS , and at least 11
ADDR_W	Integer	25	derived	Avalon word address width — row + column + bank. Do not override

4.2 Device timing

Given in **picoseconds** in the HDL and in **nanoseconds** in Platform Designer, which scales them. See section 4.5 for why.

Parameter	Default	Description
T_RC_PS	60 000	ACTIVATE to ACTIVATE, same bank
T_RAS_PS	37 000	ACTIVATE to PRECHARGE, same bank
T_RP_PS	15 000	PRECHARGE to ACTIVATE, same bank
T_RCD_PS	15 000	ACTIVATE to READ or WRITE
T_RRD_PS	14 000	ACTIVATE to ACTIVATE, different bank
T_WR_PS	14 000	Write recovery: last write data to PRECHARGE
T_MRD_PS	14 000	LOAD MODE REGISTER to next command
T_RFC_PS	60 000	AUTO REFRESH to next ACTIVATE or REFRESH
CAS_LAT	3	CAS latency, 2 or 3

4.3 Initialisation and refresh

Parameter	Default	Range	Description
T_INIT_US	100	1:10000	Power-up wait before the first command, microseconds
INIT_REFS	8	2:15	AUTO REFRESH commands during initialisation
REF_ROWS	8192	512:32768	Rows the device refreshes per period
REF_PERIOD_MS	64	1:1000	Refresh period, milliseconds
REF_MAX_PEND	8	1:8	Refreshes that may be postponed. 1 refreshes immediately

4.4 Controller options

Parameter	Default	Range	Description
CLK_KHZ	100 000	derived	Taken from the connected clock source. Every timing is converted against it
ADDR_MAP	0	0, 1	0 = compatible with the Intel core, 1 = conventional {row, bank, column}
FIFO_DEPTH	8	2, 4, 8, 16, 32	Buffered commands. Also what look-ahead looks into
LOOKAHEAD	1	0, 1	Open or close the next access's row early
RD_EXTRA_LAT	0	0:3	Extra read-capture delay. Zero is correct for a direct connection

4.5 Why picoseconds, and why not cycles

Cycle counts are deliberately not exposed. A timing parameter rounded the wrong way is silent data corruption at temperature months later, not a clean failure, and pushing that arithmetic onto every integrator guarantees someone gets it wrong.

Picoseconds rather than a `real` number of nanoseconds because **Platform Designer emits a FLOAT parameter into the generated Verilog as a quoted string** — `.T_RC_NS("60.0")`. Assigned to a `parameter real`, that string is its ASCII bytes read as a number: 60.0 arrives as 909127216.0, and a 6-cycle tRC becomes 90 million. Integers cross the tool boundary intact.

Caution

If you add a parameter to this core, check the generated wrapper and confirm it arrives unquoted.

5. Signals

20 ports.

5.1 Clock and reset

Signal	Direction	Width	Description
clk	Input	1	All logic is in this domain
reset_n	Input	1	Asynchronous assert, synchronous deassert

5.2 `s1` — Avalon-MM slave

Word-addressed, `isMemoryDevice`, variable read latency with `readdatavalid`.

Signal	Direction	Width	Avalon role
az_addr	Input	ADDR_W	address (words)
az_be_n	Input	DATA_BITS/8	byteenable_n
az_cs	Input	1	chipselct
az_data	Input	DATA_BITS	writedata
az_rd_n	Input	1	read_n
az_wr_n	Input	1	write_n
za_data	Output	DATA_BITS	readdata
za_valid	Output	1	readdatavalid
za_waitrequest	Output	1	waitrequest

5.3 `wire` — SDRAM conduit

Signal	Direction	Width
zs_addr	Output	SA_BITS
zs_ba	Output	BANK_BITS
zs_cas_n	Output	1
zs_cke	Output	1
zs_cs_n	Output	1
zs_dq	Bidir	DATA_BITS
zs_dqm	Output	DATA_BITS/8
zs_ras_n	Output	1
zs_we_n	Output	1

6. Address map

See Figure 4. At the default geometry, `ADDR_MAP 0` gives `bank = {addr[24], addr[10]}`, `row = addr[23:11]`, `column = addr[9:0]`.

The Avalon address is a **word** address, matching the core this replaces, so a system's existing address assignments carry over unchanged.

7. Performance

Measured on `benchmark/`, identical stimulus, both controllers. ISSI IS42S16320D at 100 MHz, 16-bit bus, CAS 3, 4096 operations per pattern. Theoretical peak 200 MB/s.

Pattern	Intel's core	This core	Speedup
seq write	194.3	198.3	1.02×
seq read	194.0	196.1	1.01×
seq read/write	21.9	78.9	3.6×
same-row rd/wr	21.9	78.9	3.6×
bank+row walk	21.9	65.5	3.0×
4-bank same row	21.9	194.7	8.9×
random	22.0	44.5	2.0×

0 data errors, 0 timing violations, both controllers.

Sequential streaming was already finished and is essentially unchanged. The gain is entirely in mixed and scattered traffic.

Note

These are simulation figures. The first two rows reproduce the 194 MB/s the SDRAM example measured on hardware, which is the check that the harness measures the right thing — but no figure in this table has itself been observed on a board.

7.4 Resources and f_MAX

Quartus 18.1.1 Standard, MAX 10 `10M50DAF484C7G`, Slow 1200 mV 85 °C model. Both controllers synthesised standalone under the same constraints.

	Intel's core	This core
Logic elements	353	1,193
Registers	225	779
f_MAX	115.2 MHz	101.0 MHz

The complete DE10-Lite demonstration — this controller plus sequencer, master, PLL, seven-segment displays and JTAG probes — occupies 2,805 logic elements and 1,602 registers, 6% of the device, and closes its 100 MHz constraint with 0.430 ns of setup slack. The SDRAM interface paths close with 1.996 ns.

Caution

the throughput table in 7.3 assumes 100 MHz for both controllers. Cycle counts are a property of the scheduler; megabytes per second are not. This core reaches 101 MHz and the core it replaces reaches 115, so the ratios hold at 100 MHz and below and stop holding above it. A system already running Intel's controller above 101 MHz cannot substitute this one without lowering its clock.

The critical path is the loop from the registered command-buffer head, through the scheduler's priority chain, through the pop decision, and back into that register. Buffering the FIFO output is what made 100 MHz reachable at all: with the head taken combinationally from the array, the multiplexer and the entire priority chain shared one cycle, and f_MAX was 83 MHz.

8. Verification

Flow	What it covers	Status
<code>simulation/verilator/run_sim.sh</code>	Lint of RTL, checker and model; timing-checker self-test; testbench across 13 configurations including three clock rates and both supplied parts; lint in 4 geometries; Quartus Analysis & Synthesis	22 checks, 2158 testbench assertions, passing
<code>tb/avalon_mm_sdrmm_controller_tb.sv</code>	166 checks per configuration, asserting on the command stream as well as the data	Passing
<code>tb/avalon_mm_sdrmm_controller_sva.sv</code>	Avalon protocol, command legality, DQ contention, row bookkeeping	Passing
<code>tb/sdrmm_timing_check.sv</code>	tRC, tRAS, tRP, tRCD, tRRD, tWR, tMRD, tRFC, read-to-write turnaround, refresh interval	Passing, with a 23-check threshold self-test
<code>benchmark/</code>	Throughput against the core being replaced	Passing
<code>example/de10_lite_rtl</code>	Board-level demonstration, 9 phases	58 checks passing in simulation
<code>simulation/questa/run_sim.tcl</code>	Coverage and assertion non-vacuity, same 13 configurations	22 assertion instances, none vacuous , 100% FSM state and transition
Quartus	Synthesis, fit, timing closure, bitstream	100 MHz met with 0.430 ns slack
Hardware	Retention and refresh on silicon	Not run — no board

8.1 The testbench checks the mechanism, not only the data

A controller that closed and reopened a row before every access would return correct data and pass any integrity check. So the testbench counts commands: four banks with one row open each, accessed in rotation, is asserted to cost exactly four ACTIVATES and zero PRECHARGES.

8.2 Proven by fault injection

Six faults were introduced into the controller and each was caught by the layer meant to catch it:

Fault	Caught by
Byte enables ignored	Directed test
Read/write turnaround removed	SVA
tRCD gate removed	Timing checker
ACTIVATE forgets its row	SVA
Refresh never forced under load	SVA
Read data captured a cycle late	Directed test

9. Limitations and known gaps

Limitation	Detail
Never run on hardware	The design synthesises, fits, meets timing and produces a bitstream, but has not been programmed into a part. Retention and refresh on silicon are unproven
No bursting	The slave is non-bursting; every transfer is one word. Bursts would amortise the read/write turnaround and are the main remaining performance work
No reordering	Transfers are served in order. A reorder buffer grouping same-direction accesses is the other half of that work
One chip select	Multi-device configurations are not supported
Single clock domain	No clock-crossing on the slave port
Two device presets	The DE10-Lite's IS42S16320D-7 and the DE0-Nano's IS42S16160B-7. Others need datasheet figures
f_MAX headroom is thin	101 MHz standalone against a 100 MHz target. The critical path is the loop from the registered FIFO head through the priority chain and back; shortening it further means splitting that chain across two cycles

10. Revision history

Document version	Core version	Date	Change
1.0	1.0	August 2026	First release